DPCN Group 23 — data provenance cheat sheet

Which field on the dashboard comes from which source, and the exact file/function in code. Pipeline: `run_all.py` → collect → build → dashboard → report.

1. The three sources (nothing else feeds the dashboard)

Source	What it is	What we take	What we do *not* take from it
OpenAlex api.openalex.org	Open scholarly index	Papers, authors, affiliations, citation counts, DOIs, years, journals/ISSNs, topics, references, citing works, h-index	Quartiles, CORE ranks, Journal Impact Factor
Scimago / our mapping `scimago.csv`	ISSN → Q1/Q2/Q3/Q4	Journal quartile only	Impact factor, citations
Our IF lookup `journal_if.py`	Hand list of ~2023 JCR-style IF by ISSN	IF > 5 / IF > 10 bands	Anything for ISSNs not in the dict (those stay Unlisted)

CORE A*/A/B/C is **not loaded**. OpenAlex has no CORE ranks. Conference panel is a note, not a ranking.

2. Files on disk (raw → out → screen)

File	Written by	Function / lines	Used by
`raw/institutions.json`	collect.py	`resolve()` ~L67; saved in `cached()` ~L191	build.py L31
`raw/works.json`	collect.py	`fetch_works()` L85–96 (OpenAlex /works)	build.py L31; dashboard.py L49–52 (ISSN for IF colours)
`raw/authors.json`	collect.py	`fetch_authors()` L105–114 (h-index from summary_stats)	build.py L192 `hval`
`raw/fullauth.json`	collect.py	`fetch_full_authors()` L172–187 (papers with 100+ authors)	build.py L33
`raw/refs.json`	collect.py	`fetch_light()` L149–154 (referenced works)	build.py L31, citation + Indian out-flow
`raw/citers.json`	collect.py	`fetch_citers()` L157–169 (works that cite us)	build.py L31, citation + Indian in-flow
`scimago.csv`	build_quartiles.py L544 (or fetch_scimago.py)	ISSN;Quartile;Title	build.py L40–53 `quartile()`
`journal_if.py`	hand-maintained	`IF_BY_ISSN`, `impact_factor()`, `if_band()`	dashboard.py L4, L17–40, L54–57
`out/group23_data.csv`	build.py L132–139	one row per author×paper×institute	submission file; dashboard.py L11–27 for IF counts
`out/citation_network.csv`	build.py L141–161	all citation edges incl. outside world	submission; **not** what the dashboard draws
`out/authorship_network.csv`	build.py L163–173	co-author pairs (skip >100 authors)	submission
`out/indian_connections.csv`	build.py L175–185	shared papers with Indian partners	submission; metrics connections built in same file L335–358
`out/metrics.json`	build.py L362	all KPIs, charts, networks, flow	dashboard.py L7; report.py
`out/dashboard.html`	dashboard.py L195	embeds Chart.js, vis-network, metrics JSON	the live dashboard

3. Dashboard panel → source → code

On screen	Data from	Computed in	Drawn in dashboard.py
KPI Papers	OpenAlex works, 2016–2026, attributed to an institute	build.py L198 `d["papers"]`	L116

KPI Citations	OpenAlex `cited_by_count` per work, summed	build.py L198	L116
KPI Faculty (≥5 papers)	OpenAlex authors; we apply threshold 5	build.py L18, L220–222	L116
KPI All authors	same author list, no threshold	build.py L220	L116
KPI Q1 share	Scimago quartile via ISSN; % of **known** journal quartiles	build.py L200–204	L115–116
KPI Median h-index	OpenAlex author `summary_stats.h_index`	collect.py L112; build.py L223–226	L116
KPI IF > 5 / IF > 10	`journal_if.py` + ISSN from CSV	dashboard.py L10–43 (not build.py)	L116
Publication trend	OpenAlex `publication_year`	build.py L199	chart `c_trend` L138
Quartile doughnut	Scimago Q via build.py `quartile()`	build.py L201–202	`c_q` L139
IF doughnut + high-IF table	journal_if.py	dashboard.py `_stats_from` L29–40	`c_if` L140–141; table `t_ifj` L123
Top 5 journals	OpenAlex venue `display_name`	build.py L205	`c_j` L142
h-index histogram	OpenAlex h-index, binned 0–79	build.py L224–225	`c_h` L143
Research areas	OpenAlex `primary_topic.field`	build.py L206 (top 12)	`c_f` L144
Conference CORE	would need CORE; we do not have it	build.py L207 (Unranked)	L127, L145
Faculty table	OpenAlex name, h, papers, cites; “dept” = most common field	build.py L210–222	table `t_fac` L128
Most-cited papers	OpenAlex title, journal, year, cites, doi, author count	build.py L208–209	table `t_top` L129
Citation network	OpenAlex referenced_works + our papers; IF colour from journal_if	build.py `cit_stats` L230–275 (viz = top 250); dashboard.py L53–57 colours	`n_cit` L130, L187
Authorship network	OpenAlex co-authors; Louvain via NetworkX	build.py `auth_stats` L277–305	`n_aut` L131, L187
PageRank top 10	internal citation graph	build.py L239, L262–263 NetworkX pagerank	table `t_pr` L132
Quality-weighted PageRank	same graph × Scimago Q of **cited** paper (5/3/2/1)	build.py L242–253, L264–267	table `t_wpr` L133
Betweenness / degree	co-author graph	build.py L288–300	`t_bc` `t_dg` L134
Citation in/out flow	citers.json + refs.json; Indian `country_code == IN`	build.py `flow_stats` L307–328	`c_flow` L135, L147
Top Indian institutes citing / cited	same flow counters	build.py L327	`t_ci` `t_co` L135
Connectivity + Jaccard	co-affiliations on our papers + OpenAlex subfields	build.py L335–358	table `t_cn` L136

4. OpenAlex fields we actually request

```
Works select (collect.py L87–88): id, doi, title, publication_year, type, primary_location, cited_by_count, authorships, referenced_works, primary_topic
Authors select (L109): id, display_name, summary_stats, works_count, cited_by_count → we keep h from summary_stats.h_index
```

Filter: affiliation in our 4 institute IDs, dates 2016-01-01 … 2026-12-31, types article | review | proceedings-article (collect.py L89–90).

5. What is computed by us (not downloaded as a ready-made score)

- **Faculty flag** — threshold ≥5 papers in this corpus (build.py L18). OpenAlex has no faculty field.
- **Q1 share** — Q1 count / (Q1+Q2+Q3+Q4), Unknown excluded (build.py L203–204).
- **IF bands** — ISSN lookup in journal_if.py; Unlisted if missing.
- **Internal citation graph / PageRank / weighted PageRank** — NetworkX on our papers only (build.py L230–253).
- **Louvain communities, modularity, betweenness, degree** — NetworkX (build.py L277–300). Colours on the authorship graph = these communities, not institutes.
- **In/out Indian citation flow and Jaccard** — our counters (build.py L307–358).

- **Dashboard drawing** — only ~250 most-connected nodes (build.py keep[:250] in cit_stats / auth_stats). Full counts are in the note under the graph.

6. One-line “if they ask where X is”

They ask	You say
Citation counts, years, authors, h-index, topics	OpenAlex, collected in collect.py, stored raw/*.json
Q1 / Q2 / Q3 / Q4	Scimago ISSN map scimago.csv, applied in build.py quartile()
IF > 5 and IF > 10	Our table journal_if.py, applied in dashboard.py — not OpenAlex, not Scimago
The pretty network picture	metrics.json viz (top 250) + vis-network.js; IF colours patched in dashboard.py
The 91 MB citation_network.csv	All edges including the outside world — wider than the dashboard graph
CORE conference ranks	Not in the project. OpenAlex does not provide them.

Run order: `python3 run_all.py EMAIL [API_KEY]` which calls collect.py → build.py → dashboard.py → report.py. Regenerating the HTML after IF/layout edits: `python3 dashboard.py` only (does not re-hit OpenAlex).